

YEAR 9 SCIENCE PHYSICAL SCIENCES LENSES

- Do the following activity.
- Record your results on a sheet of paper in your group.
- Take a photo and add it below this page.
- Answer the questions.

INVESTIGATION 9.9

Focusing on light

AIM To investigate the transmission of light through different lenses

Materials:

ray box kit

12 V DC power supply

sheet of white paper

ruler and fine pencil

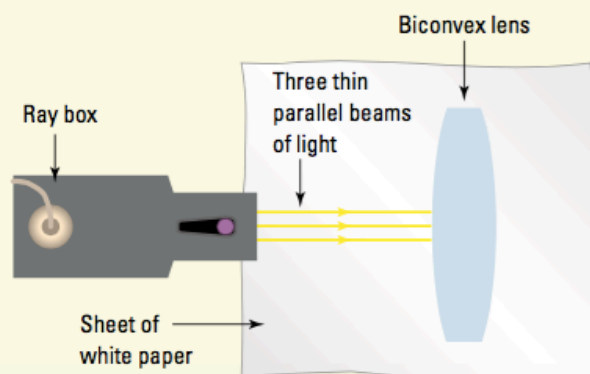
METHOD AND RESULTS

- ▶ Connect the ray box to the power supply and place it on a page of your notebook.

Part A: Biconvex lenses

- ▶ Place the thinner of the two biconvex lenses in the kit on the page and trace out its shape. Project three thin parallel beams of white light towards the lens.

- 1 Trace the paths of the light rays as they enter and emerge from the lens. Remove the lens from the paper so that you can draw the paths of the light rays through the lens.



- ▶ Replace the thin biconvex lens with a thicker one and repeat the previous steps.

Part B: Biconcave lenses

- ▶ Place the thinner of the two biconcave lenses on your notebook page and trace out its shape.

- 2 Trace the path of each of the three thin light beams as they enter and emerge from the lens. Remove the lens from the page so that you can draw the paths of the light beams through the lens.

DISCUSS AND EXPLAIN

- 3 State the focal length (distance from the focal point to the centre of the lens) for each lens.
- 4 Which of the biconvex lenses bends light more, the thin one or the thicker one?
- 5 Explain why the middle light ray does not bend.
- 6 How many times do each of the other rays bend before arriving at the focal point?
- 7 Do the diverging rays come to a focus?
- 8 Do the diverging rays appear to be coming from the same direction? Use dotted lines on your diagram to check.
- 9 Predict where the diverging rays will appear to come from if you use a thicker biconcave lens. Check your prediction with the thicker biconcave lens in the ray box kit.

Observations

- Place your photo of the results here.

Discuss and Explain

- Answer the questions here.